

Dataset Overview

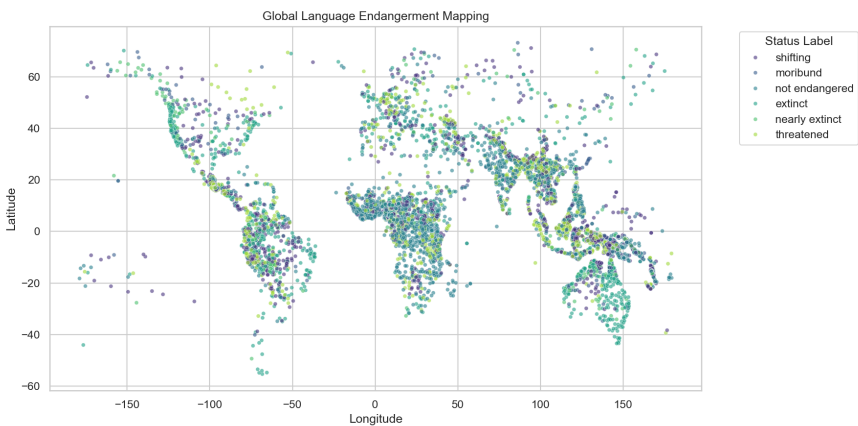
The dataset consists of 8,612 languages characterized across 12 features, capturing geographical, genealogical, and endangerment information. Representing global linguistic diversity, it records features such as macroarea, coordinates, country codes, language family, and endangerment status labels (ranging from "not endangered" to "extinct").

Data Quality Notes

- **Structural Missingness:** Family details (`family` and `family_id`) are missing for exactly 182 records, which align perfectly with the languages flagged as isolates (`is_isolate = True`).
- **Coaligned Missingness:** The status information (`status_code` and `status_label`) is missing for 474 records (5.5%), always occurring together.
- **Geographical Gaps:** Latitude and longitude coordinates are missing for 312 records (3.6%), which may affect spatial modeling and mapping.
- **ISO Code Missingness:** A total of 756 languages (8.8%) lack an ISO 639-3 code (`iso639p3code`), which is the highest rate of missingness in the dataset.

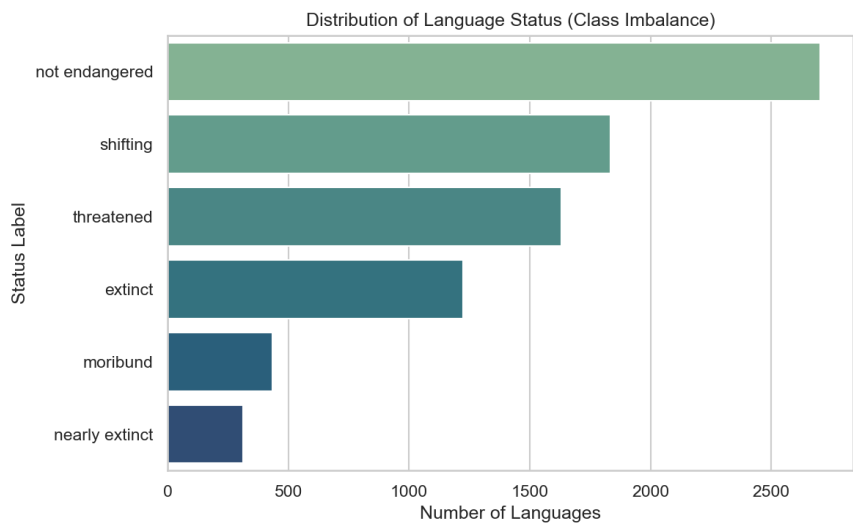
Exploratory Analysis

Chart 1: Global Language Endangerment Map



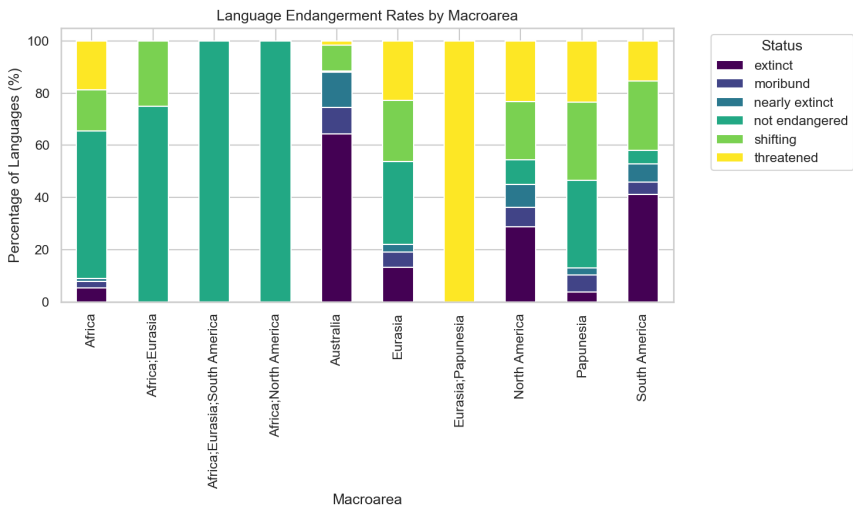
What it shows: Endangered and extinct languages are highly concentrated in North America, Australia, and parts of South America.

Chart 2: Language Status Frequency Distribution



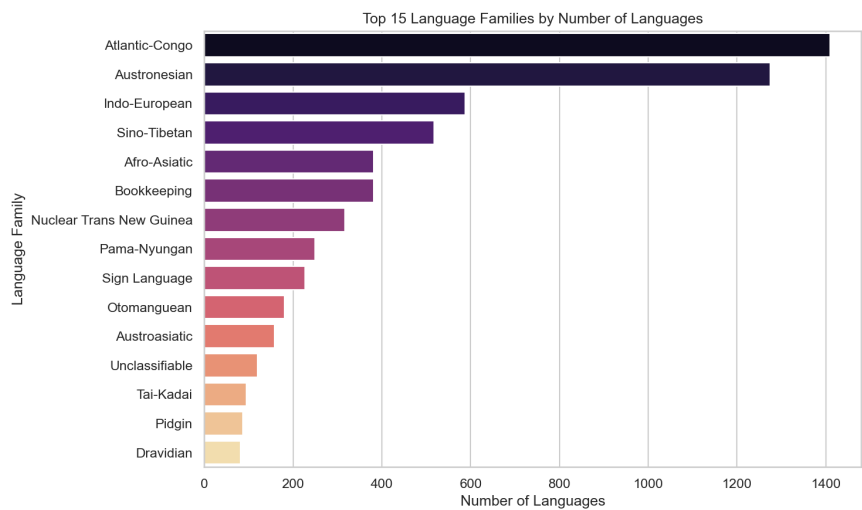
What it shows: A large portion of the world's languages are currently classified as 'not endangered', while others face varying degrees of threat.

Chart 3: Language Endangerment by Macroarea



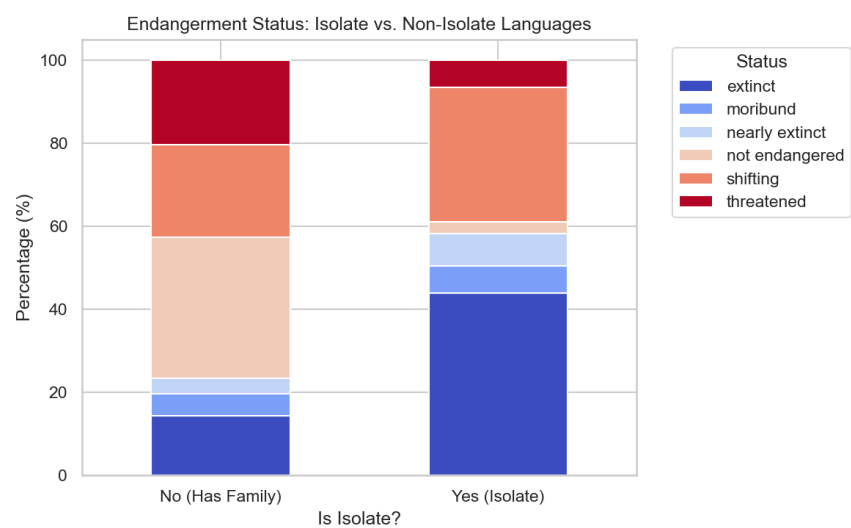
What it shows: Regions like Australia and the Americas exhibit significantly higher proportions of extinct and moribund languages compared to Africa.

Chart 4: Top 15 Largest Language Families



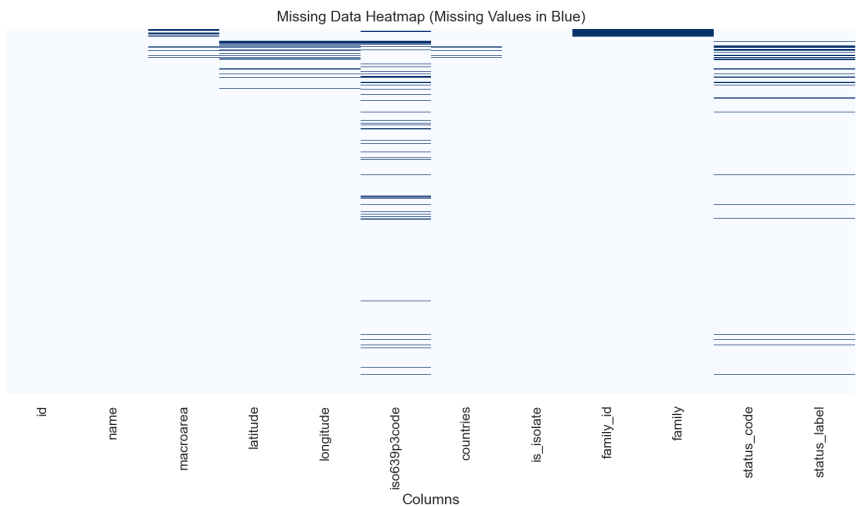
What it shows: The Atlantic-Congo and Austronesian families contain the largest number of distinct languages in the dataset.

Chart 5: Language Isolates vs. Non-Isolates Endangerment



What it shows: Language isolates exhibit a substantially higher rate of extinction and severe endangerment compared to non-isolate languages.

Chart 6: Missing Data Heatmap



What it shows: Status code and status label show perfect alignment in their missingness, confirming they are missing together.

Modeling Results

A baseline classification model was trained to predict the endangerment status label of languages. The model achieved an overall accuracy of 53.56% and a macro F1-score of 41.28%. Performance varied significantly across classes due to major class imbalances.

Status	Precision	Recall	F1-Score	Support
extinct	0.65	0.67	0.66	245
moribund	0.12	0.05	0.07	87
nearly extinct	0.26	0.18	0.21	62
not endangered	0.61	0.73	0.67	541
shifting	0.49	0.49	0.49	367
threatened	0.40	0.36	0.38	326
Overall Accuracy			0.54	1628
Macro Average	0.42	0.41	0.41	1628
Weighted Average	0.51	0.54	0.52	1628

Key Insights

Chart Insights

- **Global Language Endangerment Map**
 - Geographic plotting reveals that extinct and highly endangered languages are heavily clustered within North America, South America, and Australia.
 - High-density linguistic regions like Africa and Southeast Asia contain a large share of the dataset's viable, less-threatened languages.
 - The visualization relies on latitude and longitude coordinates that are missing for 312 languages, meaning some geographic clusters might be underrepresented.

• Language Status Frequency Distribution

- The largest single segment of languages is classified as 'not endangered', representing 2,704 of the classified languages in the dataset.
- There is a stark imbalance among the threat categories, with severe groups like 'moribund' and 'nearly extinct' having far fewer representatives than 'shifting' or 'threatened' languages.
- Because 474 languages lack status labels entirely, the visible distribution may understate the true severity of global language endangerment.

• Language Endangerment by Macroarea

- Proportional analysis reveals that Australia and the Americas face the highest rates of language extinction and severe endangerment.
- Africa displays the highest degree of language vitality, with a massive majority of its languages categorized as not endangered.
- Broad regional groupings like macroareas can obscure localized hot spots of severe endangerment within highly diverse sub-regions.

• Top 15 Largest Language Families

- The Atlantic-Congo language family is the largest in the dataset, accounting for 1,409 languages, followed closely by the Austronesian family.
- A massive portion of global linguistic diversity is concentrated in just a few large language families, while hundreds of other families remain extremely small.
- This chart focuses only on the top fifteen families, thus obscuring the status and size of hundreds of smaller, potentially highly endangered language families.

• Language Isolates vs. Non-Isolates Endangerment

- Language isolates face a disproportionately higher rate of extinction and critical endangerment than languages belonging to established families.
- Languages with family affiliations exhibit significantly higher resilience, with a larger share classified as not endangered.
- The small sample size of 182 language isolates might exaggerate the perceived correlation between linguistic isolation and threat levels.

• Missing Data Heatmap

- The status code and status label columns exhibit perfect alignment in their missingness, meaning they are always missing together.
- Structural missingness is confirmed by the fact that family data is missing exclusively for the 182 languages flagged as isolates.
- Modeling algorithms may fail or produce biased predictions if missing values in key features like coordinates or ISO codes are not handled systematically.

Overall Insights

- The dataset reveals a stark geographic divide in language endangerment, with indigenous languages in colonized regions like North America and Australia facing the highest rates of extinction.
- Linguistic isolation is a strong indicator of vulnerability, as language isolates show a much higher probability of being extinct or critically endangered than those belonging to language families.
- Severe class imbalance and a high missingness rate for status codes challenge predictive modeling, as evidenced by a low baseline classification macro F1-score of 41.28%.
- Data missingness is not entirely random, as shown by the perfect structural alignment between language isolates and missing family information.

Recommendations / Next Steps

- **Address Class Imbalance:** Use oversampling (such as SMOTE) or class-weighted loss functions to improve baseline predictions for underrepresented categories like "moribund" and "nearly extinct".
- **Impute Missing Coordinates:** Utilize external geographical registries linked by ISO codes or country fields to fill the missing spatial parameters for the 312 incomplete records.
- **Treat Isolates Distinctly:** Explicitly encode language isolates as their own categorical family class instead of leaving them as missing values (NaN) to support robust downstream model training.
- **Incorporate External Covariates:** Merge regional demographic and socioeconomic indicators (e.g., community population sizes, local education policies) to improve the model's capacity to predict language shift dynamics.